

****Note that this is a generalized list; you need to dive into the details as necessary****

Chapter 1

Taxonomic levels of organization (Domain Species) and relatedness

Formulating hypotheses

Graph interpretation

Chapter 2

Atomic structure

Valence electrons vs valence number

HONC 1234 (common bonding patterns)

Ionic bonds

Polar covalent bonds

Non-polar covalent bonds

Hydrogen bonds

Chapter 3

Polarity

Hydrogen bonds

Hydrophobic/hydrophilic

Acids and Bases

pH, [H⁺], pOH, [OH⁻] calculations

Chapter 4

Isomers

Types

identification

Functional groups